

BINLEVEL · DESIGN DECISIONS REGISTER

BinLevel design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/binlevel		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.1	2026-09-30	Amish Chadha	Register opened with the open decisions from the review note, the decision records and the build plan work

BinLevel design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

#	Decision needed	Options	Recommendation	Affects in the build	Source
1	Review the design-for-construction changes P1 to P11 (box on studs and standoffs, no tabs, M6 x 20 bolts, 55 mm box, prototyping board, printed sensor mounts, vent, plugs)	Accept; change any item	Accept	The whole build plan	BNL-DDR-003
2	Cell swap now means lowering the electronics board (four screws) after the cover is off	(a) accept, since the cell lasts about 16 years; (b) move the cell below the board, which needs a taller box	(a)	Steps 7 and 8	BNL-DDR-003, A2
3	Steel containers	(a) external lid antenna variant (about \$8, parts \$67.00, \$7.00 over the value-engineering target); (b) leave steel containers out of the first pilot	None made	Antenna and bracket; cost	BNL-DDR-001 and BNL-DDR-002, O1
4	First partner and region for co-design and a pilot	Partner and region; the region also sets the radio band (EU868, US915 or IN865)	None made	Radio module build and antenna band; whether IP67 is enough for the partner's washing practice	BNL-DDR-001 and BNL-DDR-002, O2
5	Galvanic isolation between the stainless bolts and studs and the aluminium plate	(a) anodised plate; (b) insulating washers and sleeves; (c) nothing for the prototype, decide for deployments	(c) for the prototype, (a) for deployments	Bracket plate finish	BNL-DDR-002, Consequences; review note 2026-09-25
6	Route planner and data layer for the pilot	Twinkit, a city system, or an open-source vehicle routing tool	None made	Not part of the TRL 3 build	BNL-PRC-001, open questions
7	Whether the payload also carries a rolling fill-rate estimate	Add it in the spare byte; leave it to the server	None made	Firmware only; not part of the TRL 3 build	BNL-PRC-001, open questions
8	Photoreal render choices: render pose turned 45° about the lid hinge, compact street bin as context, a device label with no BOM line	Accept each; change	Accept the pose and the street bin for renders only; add the label as a note under BOM line 1 if wanted	Renders only; the renders also need redrawing to the constructable design	Review note 2026-09-26, items 1, 2 and 7

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The stock box: 115 x 65 x 55 mm outside, a 15 mm deep cover, corner towers about 9 mm across at about 51 and 26 mm from the centre lines, and a flat base floor	The board size, the stud positions and the sealing washers depend on it	BNL-DDR-003, P1, P4 and P8
2	The transducer probe's diameter (24 mm assumed), its cable plug and its temperature rating	Sets the 25 mm hole and the collar bore; R6 is at risk on the rating	BNL-DDR-003, P5; BNL-CAL-001, section H
3	The driver board's size and mounting holes (about 41 x 28 mm assumed)	Sets the four driver holes in the electronics board and its clearance to the probe and the holder	BNL-DDR-003, P3
4	The light sensor breakout's size, holes and sensor position (13 x 18 mm, sensor underneath, assumed)	Sets the holder pocket	BNL-DDR-003, P6
5	The press-in stud's hole size, minimum sheet thickness and push-out rating in 2 mm 5052 aluminium (4.2 mm and about 900 N assumed)	The plate holes and the fixing check	BNL-CAL-001, [J3]
6	The cell holder takes a C cell with room for the strap slots	Cell retention at 20 g	BNL-CAL-001, [J2]

■ Value engineering

Value-engineering target: USD 60 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 59 (USD 1 under the target). Main cost drivers and savings worth trying:

- The largest lines are the primary lithium cell and holder (USD 9), the LoRaWAN module (USD 9), the electronics board and small parts (USD 8), the ultrasonic transducer with driver (USD 7) and the enclosure (USD 6).
- Making the design constructable (BNL-DDR-003) took the parts from USD 55 to USD 59, adding box fixings (USD 2.50) and printed sensor mounts and window (USD 1.50) and respecifying lines 1, 2, 3, 6, 7 and 9.
- The external lid antenna for steel containers would add about USD 8 and bring the estimate to USD 67, USD 7 over the target.
- Savings worth trying: confirm prices at TRL 4, and a cheaper generic box.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items D1 to D10: communal containers first, ultrasonic plus ToF, C cell, LoRaWAN on RAK3172, FieldNode and TwinKit reuse, 80 % and 1 h defaults, heat alert, lid mounting with four tamper bolts, stock IP67 box, no budget change	Amish: "i accept all your recommendations, go with them across all repos."	BNL-DDR-001 , BNL-DDR-002
2026-09-25	2 mm 5052-class aluminium bracket; R6 upper limit 70 °C with the RAK3172-T for cold sites; heat rate rule counted only above 50 °C; 11-byte payload and 5 min temperature reads; 2 h interval at SF12 only; budget stays \$60	Amish, same instruction	BNL-DDR-002 , N1 to N6
2026-09-30	Design for construction: box held on four studs and standoffs, no tabs, M6 x 20 bolts, 55 mm box, prototyping board on standoffs, printed sensor mounts, vent, plugged sensor leads	Made under Amish's 2026-09-30 instruction to make the design physically buildable ("fix the design assumptions to match and be physically feasible"); open for his review (open decision 1)	BNL-DDR-003